

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with binary code '01010011 01010100 01000001 0100100' and a central atom-like symbol with three blue dots.

DevOps

User and Group Management

Linux User

- **Root User:** has user **UID 0** and has the highest privilege in system. It can do any administrative work and can access any service. This account is intended for system administration and should be used only for this purpose. It should not be used for routine activities.
- **System User:** has user **UID from 1 to 999** and mostly run system services and processes in the background also known as non-interactive processes. System Users doesn't own home directory.
- **Regular User:** are human users created by root or another user with root privileges. Each normal user has a login shell and a home directory to store their files and typically receive UIDs from **1000** onwards.

Linux User

- `cat /etc/passwd`
 - Username
 - Password
 - User ID
 - Group ID
 - User Info
 - Home Directory
 - Login Shell

```
james@machine01 ~$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:,:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:,:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:,:/nonexistent:/usr/sbin/nologin
tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false
uuid:x:107:113:,:/run/uuid:/usr/sbin/nologin
tcpdump:x:108:114:,:/nonexistent:/usr/sbin/nologin
sshd:x:109:65534:,:/run/sshd:/usr/sbin/nologin
pollinate:x:110:1:,:/var/cache/pollinate:/bin/false
landscape:x:111:116:,:/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:112:117:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
lxd:x:999:100:,:/var/snap/lxd/common/lxd:/bin/false
james:x:1000:1000:,:/home/james:/usr/bin/zsh
bona:x:1001:1001:,:/home/bona:/bin/bash
james@machine01 ~$
```

Linux User

- `$ sudo adduser dara # Create new user named "dara"`
- `$ id dara # Show user info (user id, group id)`
- `$ sudo usermod -u 3000 dara # Change user id number`
- `$ sudo usermod -l dara new-dara # Change username`
- `$ sudo su - new-dara # Switch user to "new-dara"`
- `$ whoami # Show current username`
- `$ sudo passwd new-dara # Change user password`
- `$ sudo deluser new-dara # Delete user "new-dara"`

Note: You should add new user to sudo group "sudoers" and it will have privilege to use "sudo" command(superuser do).

Linux Group

- `cat /etc/group`
 - Group name
 - Password
 - Group ID
 - User existing

```
james@machine01 ~$ cat /etc/group | tail -n 10
tcpdump:x:114:
_ssh:x:115:
landscape:x:116:
fwupd-refresh:x:117:
admin:x:118:
netdev:x:119:
lxd:x:120:
james:x:1000:
docker:x:999:root,james,bona
bona:x:1001:
```

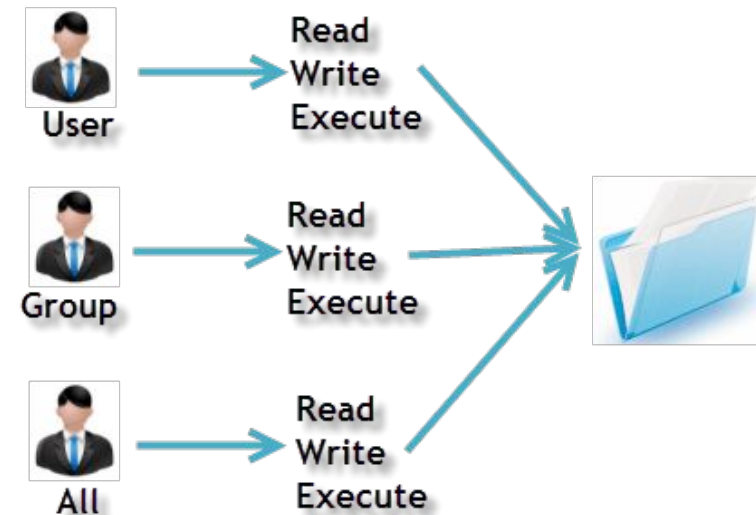
Linux Group

- `$ sudo usermod -aG sudo dara # Add user “dara” to group “docker”`
- `$ sudo addgroup dara # Create new group “dara”`
- `$ sudo groupmod -g 3333 dara # Change group id`
- `$ sudo groupmod -n new-dara dara # Change group name to “new-dara”`
- `$ groups dara # Show joining group of user “dara”`
- `$ sudo deluser dara sudo # Delete user “dara” from group “sudo”`
- `$ sudo delgroup new-dara # Delete group “new-dara”`

File Permission

- Linux is a clone of UNIX, the multi-user operating system which can be accessed by many users simultaneously. Linux can also be used in mainframes and servers without any modifications. But this raises security concerns as an unsolicited or malign user can corrupt, change or remove crucial data. For effective security, Linux divides authorization into 2 levels.
 - Ownership(User, Group, Other)
 - Permission(Read, Write, Execute)

Owners assigned Permission On Every File and Directory

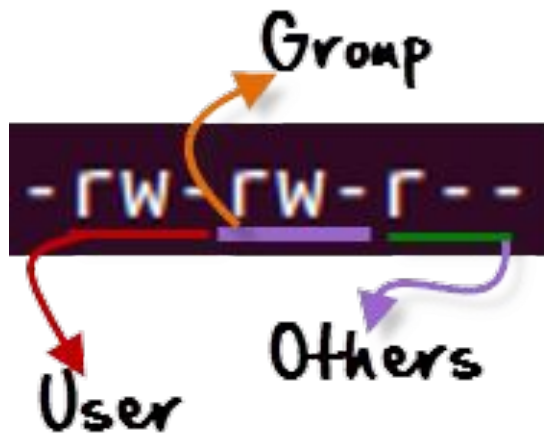


File Permission

- `$ ls -l` or `$ ll`

File type and Access Permissions.

```
home@VirtualBox: ~  
home@VirtualBox:~$ ls -l  
-rw-rw-r-- 1 home home    0 2012-08-30 19:06 My File
```



r: Read
w: Write
x: Execute

`-rw-rw-r--`

no execute
permission

File Permission

- directory, file or link

`-rw-rw-r--`



indicates
file

`d` represents directory

`drwxr-xr-x 2 ubuntu ubuntu 80 Sep 6 07:27 Desktop`

```
aaronkilik@tecmint ~ $ ln -s ~/bin/topprocs.sh topps.sh
aaronkilik@tecmint ~ $
aaronkilik@tecmint ~ $ ls -l topps.sh
lrwxrwxrwx 1 aaronkilik aaronkilik 32 Oct 7 10:39 topps.sh -> /home/aaronkilik/bin/topprocs.sh
```

→ Create Symbolic Link



Two Ways to Set Permission

- Absolute(Numeric) Mode
 - Read 4
 - Write 2
 - Execute 1
- $3 = 2 + 1$
- $5 = 4 + 1$
- $6 = 4 + 2$
- $7 = 4 + 2 + 1$

Number	Permission Type	Symbol
0	No Permission	—
1	Execute	-x
2	Write	-w-
3	Execute + Write	-wx
4	Read	r-
5	Read + Execute	r-x
6	Read +Write	rw-
7	Read + Write +Execute	rwX

Two Ways to Set Permission

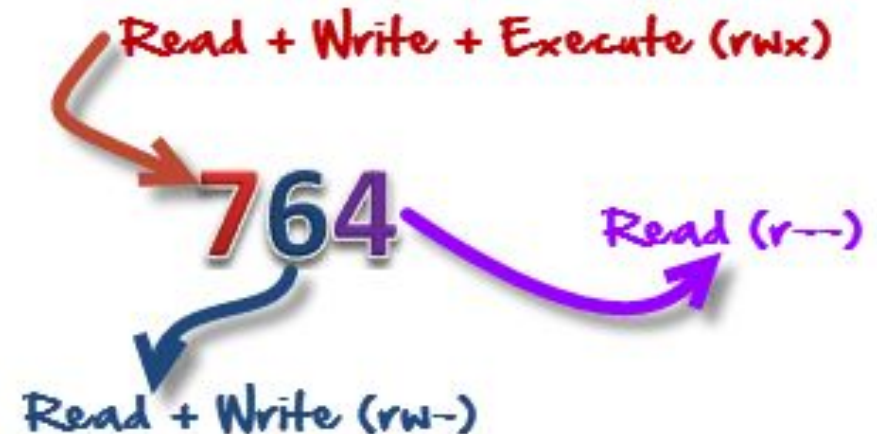
- `$ sudo chmod 777 filename.txt`

Checking Current File Permissions

```
ubuntu@ubuntu:~$ ls -l sample  
-rw-rw-r-- 1 ubuntu ubuntu 15 Sep  6 08:00 sample
```

chmod 764 and checking permissions again

```
ubuntu@ubuntu:~$ chmod 764 sample  
ubuntu@ubuntu:~$ ls -l sample  
-rwxrw-r-- 1 ubuntu ubuntu 15 Sep  6 08:00 sample
```



Two Ways to Set Permission

- Symbolic Mode

User Denotations	
u	user/owner
g	group
o	other
a	all

Operator	Description
+	Adds a permission to a file or directory
-	Removes the permission
=	Sets the permission and overrides the permissions set earlier.

Two Ways to Set Permission

- `$ sudo chmod u=rwx,g+w,o-x filename.txt` # Do not add space “u=rwx, g+w, o-x”

Current File Permissions

```
home@VirtualBox:~$ ls -l sample
-rw-rw-r-- 1 home home 55 2012-09-10 10:59 sample
```

Setting permissions to the 'other' users

```
home@VirtualBox:~$ chmod o=rwx sample
home@VirtualBox:~$ ls -l sample
-rw-rw-rwx 1 home home 55 2012-09-10 10:59 sample
```

Adding 'execute' permission to the usergroup

```
home@VirtualBox:~$ chmod g+x sample
home@VirtualBox:~$ ls -l sample
-rw-rwxrwx 1 home home 55 2012-09-10 10:59 sample
```

Removing 'read' permission for 'user'

```
home@VirtualBox:~$ chmod u-r sample
home@VirtualBox:~$ ls -l sample
--w-rwxrwx 1 home home 55 2012-09-10 10:59 sample
```


Two Ways to Set Permission

- `$ sudo chown data filename.txt`
- `$ sudo chown data:new-group filename.txt`

check the current file ownership using `ls -l`

`-rw-rw-r-- 1 root n10 18 2012-09-16 18:17 sample.txt`

change the file owner to n100 . You will need sudo
`n10@N100:~$ sudo chown n100 sample.txt`

ownership changed to n100

`-rw-rw-r-- 1 n100 n10 18 2012-09-16 18:17 sample.txt`

changing user and group to root 'chown user:group file'

`n10@N100:~$ sudo chown root:root sample.txt`

user and group ownership changed to root

`-rw-rw-r-- 1 root root 18 2012-09-16 18:17 sample.txt`

Two Ways to Set Permission

- `$ sudo chgrp new-group filename.txt`

check the current file ownership using `ls -dl`

```
guru99@VirtualBox:~$ ls -dl test1  
-rwxrwxrwx 1 root cdrom 0 Oct  6 11:27 test1
```

change the file owner to root. You will need sudo

```
guru99@VirtualBox:~$ sudo chgrp root test1
```

Group ownership changed to root

```
guru99@VirtualBox:~$ ls -dl test1  
-rwxrwxrwx 1 root root 0 Oct  6 11:27 test1
```

How to use “sudo” command without password?

- **sudo** (Superuser Do) – “Run as Administrator” in window OS is used to allow a user to use command or run program as root privilege.
- **Ways to add user as sudo**
 - Use command: `$ sudo usermod -aG sudo dara`
 - Add in sudoers file:
 - `$ sudo vim /etc/sudoers` **or** `$ sudo visudo`
 - `seng ALL=(ALL:ALL) NOPASSWD:ALL`
 - **NOPASSWD:** No need to provide password when use “sudo” command.
 - **Note:** Add to the end of the file. Otherwise it will be overwritten by sudo group or others.

How to use “sudo” command without password?

```
# User privilege specification
root    ALL=(ALL:ALL) ALL
#seng    ALL=(ALL) NOPASSWD:ALL
# Members of the admin group may gain
%admin  ALL=(ALL) ALL
```

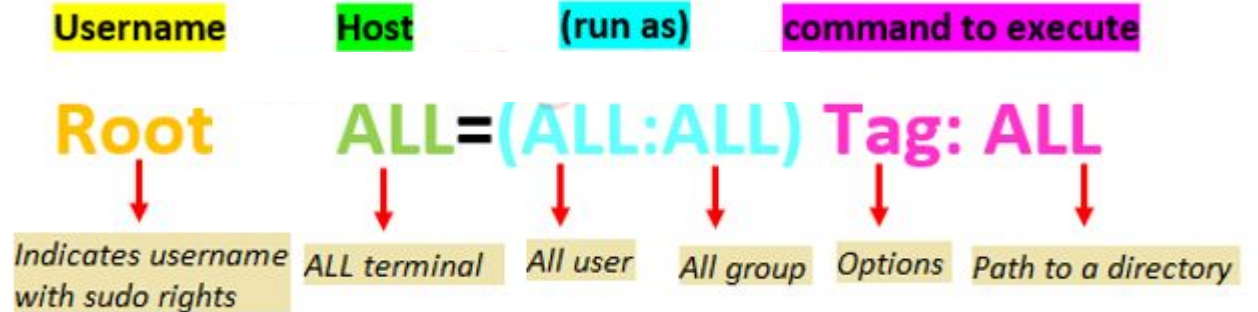
```
# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL
```

```
# See sudoers(5) for more information on "@include" directives:
```

```
@includedir /etc/sudoers.d
```

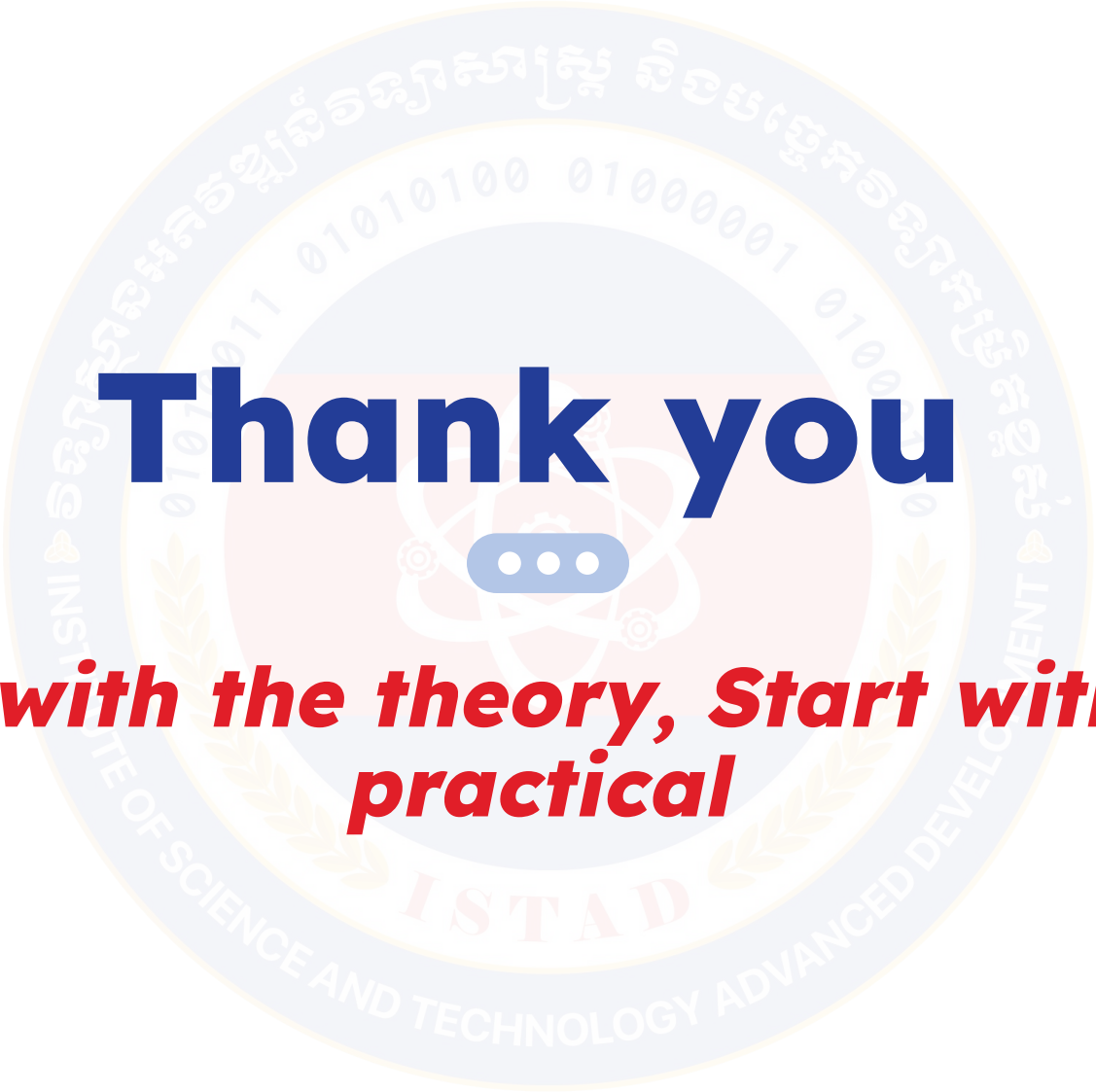
```
# Add new user privilege at the end of file.
# Otherwise it will be overwritten. %sudo, %admin...
```

```
seng ALL=(ALL:ALL) NOPASSWD:ALL
```



Example: **Raaz** **ALL=(root) NOPASSWD: /bin/cp**

```
%sudo !/bin/ls, !/bin/cd, !/bin/cat
```

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Thank you

Begin with the theory, Start with the practical